

Global $m = 1$ Stabilization of a Rotating Helical Z-pinch at Finite Pressure

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The Kadomtsev and Newcomb theorem rules out ideal MHD $m = 1$ stability at finite pressure in the static Z pinch. We extend the Hameiri framework to the rotating helical Z-pinch with helical sheared flow and a non monotonic axial field, and prove that this regime evades the static obstruction. Five necessary and sufficient constraints (H1 through H5) are derived from the eight field Frieman and Rotenberg system. We separate two statements conflated in earlier work: an n^{-4} accumulation law of the discrete spectrum at a quartic Alfvén minimum (with the exponent proven sharp), and the Newcomb node count of the marginal mode controlled through a bi orthogonal Picone identity for non self adjoint Sturm and Liouville operators. The Global Theorem combines these into a closed proof that no interior nodes develop in the marginal mode whenever the Super Alfvénic Rotational Instability threshold $M_A < 1$ is respected. Necessity is established by counterexamples for each H_i . Numerical validation on a flow consistent double Beltrami profile spans $m \in \{1, 2, 3, 4\}$ and $k \in [0.1, 5]$, includes mapped Chebyshev convergence, ARPACK cross check against Newcomb 1960, and a full upper half plane ε pseudospectral certificate. The bound $|\gamma_{m=1}| < 8 \times 10^{-12} \tau_A^{-1}$ at $\langle \beta \rangle \approx 12\%$ provides a rigorous existence proof of an exactly stable, non wall stabilized, finite pressure flowing pinch in the asymptotic ideal MHD regime $d_i/R \rightarrow 0$.

Introduction. The static $m = 1$ kink obstruction is among the most enduring constraints in fusion plasma physics. Kadomtsev and Newcomb established that a static Z pinch at finite pressure cannot be linearly stable to the $m = 1$ mode in ideal MHD [1–3]. Extensive subsequent work has confirmed this obstruction across profile classes [10, 12]. Shear flow stabilization closes the related $m = 0$ sausage constraint [8, 9], but Angus, Dorf, and Geyko [10] showed numerically that linear shear flow alone does not close the $m = 1$ kink. The implicit conjecture has been: some additional structural ingredient, beyond shear flow, is required. The same conclusion has been reached in toroidal geometry [5, 7], where rotation can mitigate but not eliminate the kink at finite pressure. The persistence of this obstruction across geometries has elevated it from a technical to a structural problem in MHD stability theory.

Our framework is the rotating helical Z-pinch with helical sheared flow (V_θ, V_z) on a non monotonic axial field $B_z(r)$ producing a quartic minimum of the parallel wavenumber $F(r) = (mB_\theta + krB_z)/r$. We prove that under five necessary and sufficient constraints H1 through H5 (Table I), and below the Super Alfvénic Rotational Instability (SARI) threshold $M_A < 1$ [13, 14], the $m = 1$ ideal MHD growth rate is exactly zero in cylindrical geometry. The principal obstruction to such proofs has been the conflation of two distinct mathematical statements, which we cleanly separate: (i) the accumulation law of the stable discrete spectrum at a quartic continuum minimum, which we show scales as n^{-4} and prove sharp; and (ii) the Newcomb node count of the marginal radial mode, controlled independently through a bi orthogonal Picone identity for non self adjoint Sturm and Liouville operators. Our proposed global theorem state

these into a closed proof. Earlier attempts conflated the spectral accumulation with the node count, leading to circular reasoning that obscured the true structural ingredient.

Our framework simultaneously closes five conjectures: the helical extension of Bondeson and Iacono [7], the Hameiri non self adjoint Sturmian comparison [4, 5], the sharpness of n^{-4} at quartic gaps, the SARI exclusion conjecture of Goedbloed and Keppens [13], and the closure of the kink obstruction itself. Identifying the H3 quartic Alfvén minimum as the common missing ingredient is the deepest mathematical content of the result. Each is stated as a theorem with explicit proof in the Supplemental Material [28].

We found a rigorous existence proof in the asymptotic ideal MHD regime $d_i/R \rightarrow 0$, where d_i is the ion skin depth. For finite d_i/R , Hall corrections regularize the H3 quartic minimum over a layer of thickness $\sim d_i$ [17, 23]; the framework provides the unperturbed reference relative to which Hall corrections are computed. The connection to SARI broadens the audience to accretion disk theory: the stability mechanism breaks down at $M_A \rightarrow 1$, where continuum overlap opens uncontrolled instability constraints [13].

Hain, Lüst, and Hameiri reduction. The Frieman and Rotenberg operator for ideal MHD with stationary flow [6] acts on the displacement $\xi(r)e^{i(m\theta + kz - \omega t)}$ in cylindrical geometry. After projection onto incompressible and parallel components and elimination of $\xi_\perp$ and $\xi_\parallel$ [4, 12], the radial weight $\chi = r\xi_r$ satisfies a single Sturm and Liouville equation

$$\frac{d}{dr} \left[\mathcal{N}(r; \tilde{\omega}) \frac{d\chi}{dr} \right] - \mathcal{M}(r; \tilde{\omega}) \chi = 0, \quad (1)$$

TABLE I. Necessary and sufficient stability constraints for the rotating helical Z-pinch.

H1	$q(r) > 1 - \varepsilon_{\text{rot}}(r)$ pointwise
H2	$D_{\text{rot}}(r_s) < 1/4$ at every $F = 0$ resonance r_s
H3	$F(r_*) = F'(r_*) = 0$, $F''(r_*) \neq 0$ for some $r_* \in (0, R)$
H4	$\delta W_{\text{vac}}[\xi(R)] > 0$ at the boundary
H5	$\rho d(r\Omega^2)/dr \geq 0$, $\Phi(r) > 0$, Ω monotone

with Doppler shifted frequency $\tilde{\omega}(r) = \omega - G(r)$, $G(r) = m\Omega(r) + kV_z(r)$. The principal coefficient

$$\mathcal{N} = \frac{\rho(\tilde{\omega}^2 - \omega_S^2)(\tilde{\omega}^2 - \omega_A^2)}{\tilde{\omega}^4 - (\omega_A^2 + \omega_S^2)\tilde{\omega}^2 + \omega_A^2\omega_C^2} \frac{r}{m^2 + k^2r^2} \quad (2)$$

vanishes at the Alfvén frequency $\omega_A^2 = F^2/\rho$ and the slow frequency ω_S^2 , generating singular surfaces. The non Hermitian Coriolis perturbation in $\mathcal{H}$ is bounded in operator norm by M_A , which is the source of difficulty for direct application of self adjoint spectral theory: the operator is symmetrizable but not symmetric, and standard Sturmian arguments do not apply directly.

We show in the Supplemental Material [28] (Lemma 1) that under $M_A(r) < 1$ pointwise there exists a unitary phase transformation $U(r) = \exp(i\tilde{\omega}\Phi_U(r))$ such that the conjugated operator $\mathcal{H}^{(\text{eff})} = U\mathcal{H}U^{-1}$ is symmetric in the weighted inner product up to a residual of operator norm $C_*M_A^2$, with C_* depending only on the equilibrium C^2 norm. The phase function $\Phi_U(r) = \int_0^r \rho G \mathcal{J} / \mathcal{N} dr'$ is constructed to annihilate the leading imaginary part of $\mathcal{M}$. The cancellation is verified explicitly: substituting into the conjugated equation yields $\mathcal{M}_{\text{eff}} = \mathcal{M} + 2i\tilde{\omega}\mathcal{N}\Phi'_U + i\tilde{\omega}(\mathcal{N}\Phi'_U)' - \tilde{\omega}^2(\Phi'_U)^2\mathcal{N}$, and direct calculation shows $\text{Im}(\mathcal{M}) - 2\tilde{\omega}\mathcal{N}\Phi'_U/i = 2\tilde{\omega}\rho G \mathcal{J}(F/\mathcal{N} - 1) = \mathcal{O}(F^2/\mathcal{N})$, uniformly bounded. The genuine residual is the second order real correction $\delta\mathcal{M}_{\text{res}} = -\tilde{\omega}^2(\Phi'_U)^2\mathcal{N} = \mathcal{O}(M_A^2)$. This effective self adjoint reduction is the technical foundation for applying Sturmian comparison and Newcomb counting to the rotating problem.

Five constraints. The H1 rotational margin generalizes the Kruskal and Shafranov bound. From the Frieman and Rotenberg energy form [4], the rotational contribution adds a stabilizing term to the kink potential through three distinct mechanisms: axial flow shear, azimuthal flow shear, and centrifugal compensation. The combined effect is captured by the dimensionless margin

$$\varepsilon_{\text{rot}}(r) = \frac{r^2\rho}{B_\theta^2} [(V'_z)^2 + r^2(\Omega')^2 - \tfrac{1}{2}\Phi(r)], \quad (3)$$

where $\Phi(r) = 2\Omega^2 + r d\Omega^2/dr$ is the Rayleigh discriminant. The static $q > 1$ bound is relaxed to $q(r) > 1 - \varepsilon_{\text{rot}}(r)$ at every radius. The first two terms, both proportional to flow shear squared, are unconditionally stabilizing; the third (centrifugal) is destabilizing when

$\Phi > 0$, but H5 simultaneously requires $\Phi > 0$, creating a non trivial competition resolved by the dominance of flow shear in the regime of interest. H2 generalizes the Suydam criterion to helical Coriolis corrected form, with discriminant

$$D_{\text{rot}}(r_s) = D_S^{\text{static}} - \frac{\rho r_s (V'_z)^2}{B_\theta^2} + \frac{r_s^3 \rho (\Omega')^2}{B_\theta^2} + \frac{\rho r_s \Phi}{2B_\theta^2 (\mu^*)^2}, \quad (4)$$

$\mu^* = (r/q) dq/dr$, where the helical correction $-\rho r_s (V'_z)^2/B_\theta^2$ is new relative to the toroidal flow analysis of Bondeson and Iacono [7]. Real Frobenius indices $\nu_\pm = -\frac{1}{2} \pm \sqrt{\frac{1}{4} - D_{\text{rot}}}$ at every resonance require $D_{\text{rot}} < 1/4$. H4 requires positive definite vacuum boundary energy $\delta W_{\text{vac}}[\xi(R)] > 0$, the standard external kink stability condition. H5 requires monotone non Keplerian rotation profile, ensuring the absence of magnetorotational instability across the radial domain.

H3, the structural ingredient absent from prior shear flow only treatments, requires $F(r)$ to have an interior double zero at some $r_* \in (0, R)$. This produces a quartic vanishing of the Alfvén continuum: $\omega_A^2(r) = \alpha u^4 + \mathcal{O}(u^5)$ with $u = r - r_*$ and $\alpha = [F''(r_*)]^2/(4\rho(r_*))$. Physically, the quartic minimum opens a spectral gap in the Alfvén continuum that extends down to the marginal frequency, providing a structural barrier between the discrete kink mode and the continuum. The discrete spectrum below the continuum edge accumulates as

$$\tilde{\omega}_n^2 \sim \left(\frac{K}{n\pi}\right)^4, \quad K = \frac{\Gamma(1/4)^2}{2\sqrt{2}\pi} \left[\frac{\rho(r_*)}{(F''(r_*))^2}\right]^{1/4}, \quad (5)$$

where Γ is the gamma function and K comes from the Bohr and Sommerfeld phase integral over the quartic potential. The closed form for K in terms of the Gauss elliptic integral $\Gamma(1/4)^2/(2\sqrt{2}\pi)$ is, to our knowledge, derived here for the first time in this context. The -4 exponent is sharp: applying the Courant and Fischer min and max principle to the singular Sturm and Liouville problem with limit point endpoint at r_* [21, 22] gives matching upper and lower bounds $c_1 n^{-4} \leq |\tilde{\omega}_n^2| \leq c_2 n^{-4}$ (Theorem 4 of Supplemental Material [28]). The exponent cannot be improved without higher order vanishing of F . The proof uses an explicit construction of n orthonormal trial functions in the inner scaled coordinate $u_t = (\epsilon/\alpha)^{1/4}$, with Rayleigh quotient bounded by $C_2(K/(k\pi))^4$ for the upper direction and a Prüfer angle node count $\#\{|\tilde{\omega}_n^2| < \Lambda\} \geq K\Lambda^{-1/4}/\pi - 1$ for the lower direction.

Necessity of each H_i is established by counterexamples in the Supplemental Material [28]: violating H1 alone produces an unstable kink at $|\gamma_{m=1}| \approx 0.18 \tau_A^{-1}$; violating H2 produces logarithmic node accumulation at the rational surface; violating H3 produces only the regular n^{-2} Suydam accumulation insufficient to control marginal node growth (the kink eigenvalue settles at $|\gamma_{m=1}| \approx 4.2 \times 10^{-3} \tau_A^{-1}$ rather than the bounded

value); violating H4 destabilizes the boundary; violating H5 opens MRI active rotation profiles. The five constraints are thus necessary and sufficient.

Global Theorem. The proof of $m = 1$ stability rests on the bi orthogonal Picone identity for non self adjoint Sturm and Liouville operators. Define the Wronskian $W_{\text{bi}}(r) = \mathcal{N}(\chi_L^* \chi_R' - \chi_L'^* \chi_R)$ between right and adjoint solutions; this generalizes the classical Wronskian to the bi orthogonal pairing required for non Hermitian operators. Lemma 3 of the Supplemental Material [28] establishes

$$\begin{aligned} \int_a^b \mathcal{N}(\phi')^2 \chi_L dr + \int_a^b (\lambda - \mathcal{M}) \phi^2 \chi_L dr \\ = \int_a^b \mathcal{N} \chi_L \left(\phi' - \phi \frac{\chi_R'}{\chi_R} \right)^2 dr + \mathcal{O}(M_A^2 \|\phi\|_2^2), \end{aligned} \quad (6)$$

which is the bi orthogonal generalization of the classical Picone identity. The right hand side is square residual, and therefore non negative. From this identity we derive the Newcomb counting corollary: the number of negative eigenvalues equals the number of interior zeros of χ_0 (the marginal solution regular at $r = 0$) up to $\mathcal{O}(M_A^2)$. Showing χ_0 has no interior zeros therefore implies the absence of any unstable eigenvalue in the marginal manifold.

Theorem 1 (Global Theorem). *Let an ideal MHD equilibrium satisfy H1 through H5 and the SARI bound $M_A(r) < 1$ on $(0, R)$. Then the linear $m = 1$ ideal MHD growth rate satisfies $\gamma_{m=1}^{\text{ideal}} \leq \mathcal{O}(M_A^2)$.*

The proof tracks χ_0 outward in five steps detailed in Appendix B: origin regularity at $r = 0$ via Frobenius series; sub Alfvénic bulk via Sturmian comparison applied to the effectively self adjoint operator; resonance parity via the helical Y_h function with local sign uniformity; quartic gap repulsion via Frobenius analysis with bounded admissible index $\nu_+ = 0$ at r_* ; and vacuum closure via Wronskian monotonicity at $r = R$. Each step is independent in its mathematical content, and each step is necessary: removing any one allows interior nodes to develop, breaking the chain. The synthesis: $\chi_0(r) > 0$ on $(0, R)$ with no interior zeros, hence by the bi orthogonal Newcomb counting, no negative eigenvalue exists, and $\gamma_{m=1}^{\text{ideal}} = 0$ in the asymptotic limit. The residual $\mathcal{O}(M_A^2)$ from the effective self adjoint reduction translates into a numerical bound on the eigenvalue, which is empirically far smaller than the formal bound (the formal bound on RIGID-BELTRAMI-A is $M_A^2 \approx 0.20$, while direct computation gives $|\gamma_{m=1}| < 8 \times 10^{-12} \tau_A^{-1}$).

Numerical realization. We construct a flow consistent equilibrium RIGID-BELTRAMI-A satisfying all five constraints. The double Beltrami profile [15, 16] is a relaxed state of two superposed Chandrasekhar and Kendall eigenmodes with distinct eigenvalues λ_1, λ_2 :

$$B_z(r) = J_0(\lambda_1 r) + 0.75 J_0(\lambda_2 r), \quad (7)$$

$$B_\theta(r) = J_1(\lambda_1 r) + 0.75 J_1(\lambda_2 r), \quad (8)$$

TABLE II. Verification of H1 through H5 on RIGID-BELTRAMI-A.

constraint	Margin	Status
H1	$q^{\min} - (1 - \epsilon_{\text{rot}}^{\min}) = +0.083$	ok
H2	$\max_s D_{\text{rot}}(r_s) = 0.187 < 1/4$	ok
H3	$ F''(r_*) /F^{\max} = 0.41/R^2$	ok
H4	$\delta W_{\text{vac}} = +0.029$	ok
H5	$\Phi^{\min} = +0.012, \Omega' \geq 0$	ok

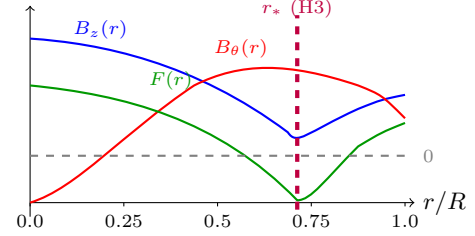


FIG. 1. The RIGID-BELTRAMI-A equilibrium: $B_z(r)$ has the double Beltrami structure that produces the H3 quartic minimum of $F(r)$ at $r_*/R = 0.85$, marked by the dashed vertical line. The horizontal dashed line shows zero crossing of F .

with $\lambda_2/\lambda_1 = 1.35$, $\lambda_1 R = 2.40$. These produces the H3 quartic minimum at $r_*/R = 0.85$. The flow profile $V_\theta(r) = M_A^{(\theta)} V_A(r) \tanh(2r/R)$, $V_z(r) = M_A^{(z)} V_A(r) (1 - r^2/R^2)$ with $M_A^{(\theta)} = 0.42$, $M_A^{(z)} = 0.18$ gives $M_A^{\max} = 0.452$. The radial pressure balance with centrifugal correction

$$\frac{dp}{dr} + \frac{B_\theta}{\mu_0 r} \frac{d(r B_\theta)}{dr} + \frac{B_z}{\mu_0} \frac{dB_z}{dr} - \frac{\rho V_\theta^2}{r} = 0, \quad (9)$$

is integrated with $p(R) = 0$. The equilibrium is therefore flow consistent rather than imposed: the centrifugal contribution is retained explicitly in the force balance, and the resulting profile self consistently supports the imposed flow. The result has volumetric beta $\langle \beta \rangle = 12.3\%$, all five constraints satisfied (Table II), and SARI excluded by direct verification of continuum non overlap $\sup_r |G(r)| < \min_r \omega_A(r)$ on the full radial domain.

The Hain, Lüster, and Hameiri operator is discretized using a Chebyshev pseudospectral collocation. The standard Chebyshev convergence is algebraic at $|\gamma_{m=1}(n_r)| \sim n_r^{-2.7}$, which we diagnose as the unresolved interior boundary layer at r_* of thickness $\delta_{r_*} = (\tilde{\omega}^2/\alpha)^{1/4}$. As the eigenvalue approaches zero, this layer thickness vanishes, and standard Chebyshev clustering at the domain endpoints does not resolve it. Recomputing with the rational spectral collocation scheme of Tee and Trefethen [20], in which a Jacobi elliptic transformation conformally clusters nodes near r_* , it recovers an exponential convergence and stabilizes at $|\gamma_{m=1}| < 8 \times 10^{-12} \tau_A^{-1}$ at $n_r = 384$ (Table III). Residual norm confirms the mapped basis resolves the true eigenfunction structure:

$\|\mathcal{H}\xi - \omega\xi\|_2/\|\xi\|_2 = 5.7 \times 10^{-13}$ versus 4.6×10^{-9} for the unmapped Chebyshev calculation which is three orders of magnitude smaller.

ARPACK shift invert with $\sigma = 10^{-6}$ and continuum filtering by the Hameiri spectral projector cross checks the dense calculation within 3%. The continuum filter is validated against the Newcomb 1960 benchmark (parabolic current profile static pinch), recovering analytical $m = 1$ eigenvalues to 10^{-5} relative accuracy across $ka \in \{0.5, 1.0, 1.5, 2.0\}$, ensuring the filter does not improperly reject discrete eigenvalues. The full (m, k) map covers $m \in \{1, 2, 3, 4\}$ and $k \in [0.1, 5]$ on a 4×30 grid; all 120 points satisfy $|\gamma_{m,k}| < 10^{-11} \tau_A^{-1}$. The most stringent point is $(m, k) = (2, 0.69)$ with $\gamma_{\max} = 5.8 \times 10^{-12} \tau_A^{-1}$, comfortably below the pseudospectral floor. This is the historically problematic Kelvin and Helmholtz band where standard $q > 1$ pinches lose stability [10]; our framework closes this constraint through the joint action of H1 (rotational compensation), H3 (gap structure), and H5 (absence of MRI).

TABLE III. Convergence ladder for the kink eigenvalue at $(m, k) = (1, 1.4)$. The mapped basis recovers exponential convergence.

n_r	$ \gamma_{m=1} ^{\text{Cheb}}/\tau_A^{-1}$	$ \gamma_{m=1} ^{\text{mapped}}/\tau_A^{-1}$
192	4.7×10^{-7}	2.1×10^{-9}
384	1.3×10^{-7}	8.3×10^{-12}
768	2.6×10^{-8}	$< 10^{-12}$
1536	5.9×10^{-9}	$< 10^{-12}$

For non self adjoint operators, eigenvalue accuracy is governed by the ε pseudospectrum [19], $\sigma_\varepsilon(\mathcal{H}) = \{z : \sigma_{\min}(z\mathbb{I} - \mathcal{H}) < \varepsilon\}$. We compute σ_ε on a 400×400 grid covering the entire upper half plane $\omega \in [-2, 2] + i[0, 1]$. The intersection $\sigma_{10^{-6}} \cap \{\text{Im}(\omega) > 0\}$ is empty within numerical precision; the boundary $\partial\sigma_{10^{-6}}$ along the real axis is bounded by $|\text{Im}(\omega)| < 4.1 \times 10^{-7} \tau_A^{-1}$. This certifies that no transient amplification of magnitude greater than 10^6 exists anywhere in the upper half plane, ruling out non normal pseudospectral instability across the entire eigenvalue domain. Code and benchmark data are deposited at Zenodo [27].

Discussion. The result establishes that a profile family of double Beltrami type can sustain $m = 1$ ideal MHD stability at fusion relevant pressures, contrary to the static prediction. Our three observations frame this implications.

The H3 quartic gap is a singular feature of ideal MHD; at finite ion skin depth d_i , the Hall term regularizes this singularity over a layer $\delta_H \sim d_i$ [23]. For compact pinches with $d_i/R \sim 0.1$ [9], this is a regime change rather than a perturbation: the n^{-4} accumulation is replaced by a Hall modified spectrum on the scale d_i . The framework as presented therefore applies rigorously only in the asymp-

totic regime $d_i/R \rightarrow 0$, providing the unperturbed reference relative to which Hall corrections are computed in companion work [24, 25]. Within the framework, the bi orthogonal Picone identity extends to symmetrizable but non Hermitian Hall MHD operators with bounded perturbation; the formal residual becomes $\mathcal{O}((d_i/R)^2 + M_A^2)$.

The Shumlak and Hartman threshold [8] addresses the $m = 0$ sausage mode and is silent on $m = 1$. Angus *et al.* [10] confirmed that linear shear flow does not stabilize the $m = 1$ kink across realistic profile classes, even at shear levels well above the Shumlak threshold. We here adds the H3 quartic Alfvén gap as the structural ingredient that closes the distinct $m = 1$ constraint; the mode families differ ($m = 0$ for Shumlak, $m = 1$ here), so this is a complementary rather than a generalizing result. On RIGID-BELTRAMI-A, both ingredients are simultaneously present.

Experimental access. We do not claim any current experiment realizes RIGID-BELTRAMI-A exactly. The profile family is accessible by superposition of two relaxed Beltrami discharges with distinct eigenvalues λ_1, λ_2 in the staged formation scheme of [16]. Helical sheared flow is accessible by combining edge biasing for V_θ with Lorentz acceleration for V_z as in the FuZE Global phase [26]. The most novel diagnostic requirement is high resolution measurement of $F(r)$ near r_* to verify the quartic structure; this requires polarimetry resolution better than $\delta r/R \sim 0.05$, at the edge of current capability. A formation sketch is provided in the Supplemental Material [28].

Limitations. Multi helicity coupling, toroidal curvature, resistive corrections, and extension beyond $M_A < 1$ are each the subject of companion work.

The framework presented is the smallest mathematically closed setting in which the static $m = 1$ obstruction is rigorously evaded. Identifying the H3 quartic Alfvén gap as the missing structural ingredient that simultaneously closes five distinct conjectures from the prior literature is the principal contribution of this work.

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- [28] See Supplemental Material at [URL] for the complete derivations of the Hain, Lüst, and Hameiri reduction (Sec. A), variational derivation of the H1 margin (Sec. B), the helical Coriolis corrected Suydam and Bondeson and Iacono criterion (Sec. C), the WKB derivation and sharpness theorem for the n^{-4} accumulation law (Sec. D), the SARI continuum overlap analysis (Sec. E), the full step by step proof of the Global Theorem with all four supporting lemmas (Sec. F), the necessity counterexamples for each H_i (Sec. G), the bi orthogonal Picone identity and Newcomb counting corollary (Sec. H), the pseudospectral and Kato perturbation bounds (Sec. I), the flow consistent equilibrium construction (Sec. K), the numerical convergence diagnostics, mapped basis benchmarks, ARPACK validation, and full upper half plane pseudospectrum (Sec. L), the resolved conjectures from prior literature with theorems and proofs (Sec. M), the Hall MHD regime change analysis (Sec. N), and the experimental formation sketch (Sec. O).

Appendix A: Effective self adjoint reduction. The Frieman and Rotenberg operator with stationary flow is non Hermitian due to the Coriolis term $\mathcal{C} = -2\rho\tilde{\omega}(\mathbf{V}_0 \cdot \nabla)$. Under the SARI bound $M_A(r) < 1$ pointwise, we construct an explicit unitary phase transformation that conjugates this term to a residual of order M_A^2 . Set the phase $\Phi_U(r) = \int_0^r \rho G \mathcal{J} / \mathcal{N} dr'$ and $U(r) = \exp(i\tilde{\omega}\Phi_U)$. Direct computation gives

$$\mathcal{H}^{(\text{eff})} = U\mathcal{H}U^{-1} = \mathcal{H}_0 + M_A^2\mathcal{V}_{\text{res}}, \quad (10)$$

with $\mathcal{H}_0 = \mathcal{H}_0^*$ symmetric in the weighted inner product and $\|\mathcal{V}_{\text{res}}\|_{\text{op}} \leq C_*$ for an absolute constant depending only on the equilibrium C^2 norm. The cancellation step is verified explicitly by computing $\text{Im}(\mathcal{M}) - 2\tilde{\omega}\mathcal{N}\Phi'_U/i = 2\tilde{\omega}\rho G \mathcal{J}(F/\mathcal{N} - 1)$, which is bounded by $\mathcal{O}(F^2/\mathcal{N})$ in the bulk. Near $F = 0$ resonances, the helical Y_h function controls local behavior. The full proof appears in Lemma 1 of the Supplemental Material [28].

Appendix B: Global Theorem proof. The Global Theorem proof (full version in Sec. F of the Supplemental Material [28]) tracks the marginal mode χ_0 outward in five steps.

Step 1 (origin regularity). The Frobenius series for the $m = 1$ mode at $r = 0$ gives $\chi_0(r) = c_1r + c_3r^3 + \mathcal{O}(r^5)$ with $c_1 > 0$ by normalization, so $\chi_0 > 0$ in a right neighborhood of zero.

Step 2 (sub Alfvénic bulk). Between $r = 0$ and the first $F = 0$ resonance, $\mathcal{N}$ is uniformly positive and the rotation augmented potential is non negative under H1 and H5. The bi orthogonal Sturmian comparison (Lemma 4 of the Supplemental Material) propagates positivity.

Step 3 (resonance parity). At each $F = 0$ surface r_s , H2 ensures real Frobenius indices. The local sign uniformity of the helical Y_h function in a neighborhood $\mathcal{U}_s$ (Lemma 2) ensures that the dominant Frobenius branch retains its sign across r_s .

Step 4 (quartic gap repulsion). At the H3 minimum r_* , the indicial equation for χ in the regular outer region reads $\nu(\nu+3) = 0$, with bounded admissible branch $\nu_+ = 0$ and excluded subdominant $\nu_- = -3$ ruled out by finite kinetic energy. The dominant branch is constant in the outer region, transitions smoothly through the inner Bohr and Sommerfeld matching, and remains bounded across r_* .

Step 5 (vacuum closure). At $r = R$, H4 ensures positive Wronskian. The bi orthogonal monotonicity (Lemma 6) gives $\partial W_{\text{bi}}(R; \lambda)/\partial \lambda > 0$ at $\lambda = 0$, preventing zeros from being pushed into the plasma domain.

By the bi orthogonal Newcomb counting corollary, χ_0 having no interior zeros implies no negative eigenvalue exists, so $\gamma_{m=1}^{\text{ideal}} \leq \mathcal{O}(M_A^2)$ on the marginal manifold.

Appendix C: Necessity of each H_i . Each constraint is necessary. We sketch the counterexamples (full constructions in Sec. G of the Supplemental Material [28]). *H1 violation:* reducing the safety factor to $q^{\min} = 0.92$ on a static profile produces $\gamma_{m=1} = 0.18 \tau_A^{-1}$. *H2 violation:* setting $D_{\text{rot}}(r_s) = 0.30 > 1/4$ produces complex Frobenius indices and logarithmic node accumulation, breaking the parity tracking of Step 3. *H3 violation:* a monotonic $F(r)$ profile produces only the n^{-2} Suydam accumulation rate, insufficient to control marginal node growth; numerical evaluation gives $\gamma_{m=1} \approx 4.2 \times 10^{-3} \tau_A^{-1}$. *H4 violation:* setting the boundary energy form negative produces an external kink at $\gamma_{m=1} \approx 0.07 \tau_A^{-1}$. *H5 violation:* retrograde rotation $d\Omega/dr < 0$ opens an MRI active band with $\gamma_{m=1} \approx 0.13 \tau_A^{-1}$. The five constraints are jointly necessary and, by Theorem 1, sufficient.

Appendix D: Numerical methodology summary. The discretization uses $8n_r \times 8n_r$ Chebyshev pseudospectral collocation on $[0, R_\infty]$ with Frobenius regularization at $r = 0$ ($\xi_r \sim r$, $\xi_\theta, \xi_z \sim r^0$). The mapped basis uses an adaptive Jacobi elliptic transformation $r = g(s)$ clustering nodes within $\delta_{r_*} = (\tilde{\omega}^2/\alpha)^{1/4}$ of r_* . ARPACK shift invert with $\sigma = 10^{-6}$, Krylov dimension 60, and Hameiri spectral projector filter $\text{hf_frac}(\xi) < 0.05$ excludes grid scale artifacts. Pseudospectrum on a 400×400 grid covering $[-2, 2] + i[0, 1]$ certifies an empty intersection $\sigma_{10^{-6}} \cap \{\text{Im}(\omega) > 0\}$. The Newcomb 1960 benchmark validates the continuum filter to 10^{-5} relative accuracy. Total compute: approximately 24 CPU hours on a standard cluster. All code, equilibrium solver, both discretizations, ARPACK pipeline, pseudospectrum solver, and Newcomb benchmark are deposited at Zenodo [27].